

Maxwell's Equations

Chapter 32, R.H.W 10th Edition

Maxwell's Equations; Magnetism of Matter.

Key ideas: The simplest magnetic structures are magnetic dipoles. Magnetic monopoles do not exist.

Gauss Law for magnetic fields

Gauss law for magnetic fields is a formal way of saying that magnetic monopoles do not exist.

The law asserts that the net magnetic flux ϕ_B through any closed Gaussian surface is zero.

$$\phi_B = \oint \vec{B} \cdot d\vec{A} = 0 \quad (\text{Gauss law for magnetic field})$$

contrast with electric field is

$$\phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{enc}}}{\epsilon_0} \quad (\text{Gauss' law for electric field})$$

Induced Magnetic fields

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d\phi_B}{dt} \quad (\text{Faraday's law of induction}) \quad (1)$$

$E \rightarrow$ Electric field induced along a closed loop by the changing magnetic flux ϕ_B encircled by that loop.

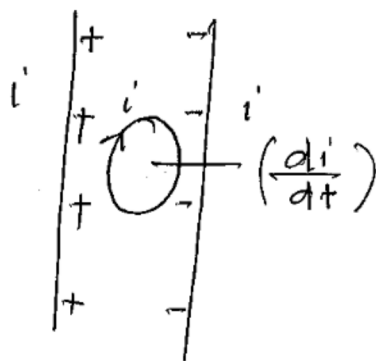
Question Does changing electric flux induced a magnetic field?

Answer is yes by Maxwell's and we call it Maxwell's law of induction —

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$$

$$\oint \vec{E} \cdot d\vec{s} = - \frac{d\phi_B}{dt} \quad \text{Faraday's law of induction}$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt} \quad \text{Maxwell's law of induction}$$



Ampere - Maxwell Law

Ampere's law $\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$
 Current encircled by the loop.

Thus two equations specify the magnetic field $\vec{B}$ produced by means other than a magnetic material, give the field in exactly the same form.

Combining the two equations,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{enc} \quad (\text{Ampere-Maxwell law})$$

$\parallel$ $\parallel$
 0 0 where there is ~~no~~ a variation of current.
 When there is not variation on current.

Displacement Current

Right hand side product $\epsilon_0 \frac{d\Phi_E}{dt}$

must have the dimension of a current compare with $\mu_0 i_{enc}$. In fact, that product has been treated as being a fictitious current called the displacement current i_d .

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt} \quad (\text{displacement current}) \Rightarrow \epsilon_0 \frac{d(EA)}{dt} = \epsilon_0 A \frac{dE}{dt}$$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} + \mu_0 i_d \quad (\text{Ampere-Maxwell's law})$$

fixed circuit

Sample problem 32.01

A parallel-plate capacitor with circular plates of radius R is being charged



(a) Derive an expression for the magnetic field at radius r for the case $r \leq R$.

